Do Now · retrieval

Unfamiliar algorithm design and evaluation

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What is a precondition?

Q2

Why write edge-case tests?

Q3

What does a loop invariant express?

Q4

C# array [4,1,4,2]: compute sum at indexes 0 and 2, and number of distinct values.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

State the precondition — The input must be sorted; equal values then appear consecutively.

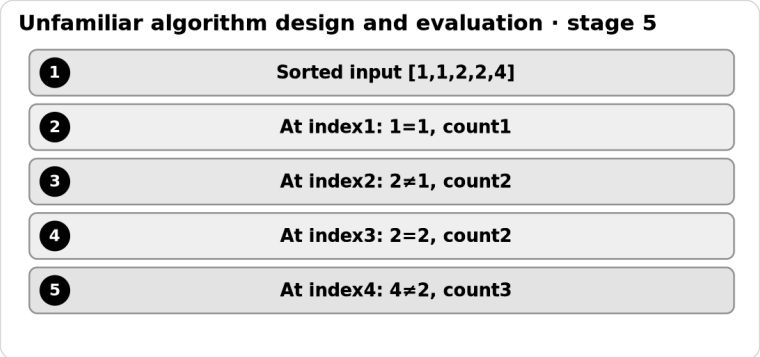
Core idea 2

Trace the invariant — Start with one distinct value for a nonempty array. Each change from previous value adds one.

Core idea 3

Evaluate and test — The algorithm scans once O(n) time, O(1) extra space; test empty, all equal and all distinct inputs.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

invariant — Condition maintained throughout a loop.

precondition — Assumption required for correctness.

boundary test — Tests around value/size limits.

time complexity — Growth in work as input increases.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

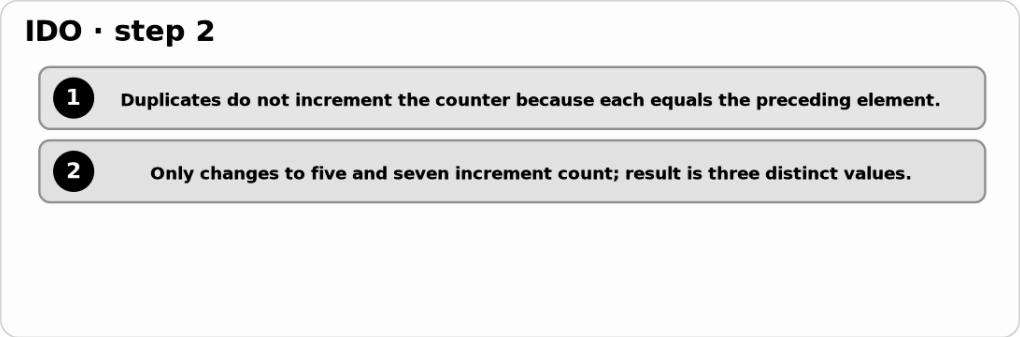
Trace distinct count in [2,2,5,5,7,7,7].

Follow the worked method

1. Predict first — Duplicates do not increment the counter because each equals the preceding element.

2. Explain the method — Only changes to five and seven increment count; result is three distinct values.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict first

Three initial ones produce one distinct group; 3 and 4 add two groups.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Trace DistinctCount on [3,3,3,6,8,8]. Record count after each index. [6]

Question 2

Describe a correct set of test cases for the algorithm and their expected outputs. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Explain why the algorithm may fail on unsorted [2,1,2]. Give a repair and complexity consequence. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

The distinct-count algorithm is valid for unsorted inputs without modification.

Use key vocabulary from memory

1. Explain invariant in your own words.

2. Explain precondition in your own words.

3. Explain boundary test in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.